

SERGIO FALCON

FULL STACK DEVELOPER

 KNOXVILLE, UNITED STATES

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Summary

Experienced full stack developer adept in all stages of web development bringing forth expertise in design, integration, testing and maintenance of web systems. Proficient in an assortment of technologies including JavaScript, .NET, C#, Python and Rust. Able to effectively self-manage during independent projects, as well as collaborate in a team setting.

EMPLOYMENT HISTORY

Software Developer at Aisin, Clinton, TN

May 2024 — Present

- Designed and built a modern TPDA system to replace an outdated legacy application, eliminating recurring inefficiencies and saving the company hundreds of thousands of dollars
- Engineered a role-based inventory and checkout system managing 43,000+ items across multiple locations using Vue and .NET Core (C#)
- Integrated LIFX smart lights into a synchronized factory-floor system that dynamically displays individual machine statuses (running smoothly, out of order, requires assistance) and alerts the appropriate workers
- Built the DCM Monitor application (Svelte and Python FastAPI) that reviews die-cast machine camera images captured every few seconds, detecting slag/debris overflow and alerting the appropriate personnel
- Established a data collection and labeling pipeline to train a YOLO computer-vision model for automated detection of slag buildup in die-cast machines
- Created a cleanroom sensor monitoring application that displays real-time sensor data on intuitive gauges, enhancing operational oversight using Svelte and Axum (Rust)
- Set up and deployed a blue-green deployment strategy to minimize application downtime during releases
- Developed robust notification background service using Quartz.NET and MySQL for scheduled delivery and automated retry, ensuring high reliability and message persistence

Full Stack Developer at RegScale, Knoxville, TN

November 2022 — November 2023

- Developed on a full stack web application using Angular, .NET Core, C#, Python, GraphQL, SQL, Docker, Bootstrap, and Kendo UI
- Developed and implemented features for document template management, specifically for substituting parameters in a word document, along with creating, saving, updating, and deleting these documents
- Created a user interface for a user to control, monitor, and manage a Job Scheduler for the RegScale CLI
- Led the initiative to implement 508 Web Accessibility features across the application, ensuring compliance with federal standards and enhancing usability
- Increased Web Accessibility coverage to 95%, reducing the number of common accessibility issues by 80%

Full Stack Developer at GovReady, Remote

September 2021 — November 2022

- Developed on a full stack web application React, Django, DjangoRestFramework, Jinja, Redux, PostgreSQL, Docker, Bootstrap, Material UI, Cypress, and Selene
- Worked alongside a team on the development of a real-time collaboration document editor, integrating a suite of components for templating, polling, and enhanced interactivity
- Developed React components to leverage backend RESTful APIs to populate client data and allow users to manage data as needed
- Implemented systems that adhered to OSCAL standards
- Developed on a full stack web application using React, Django, PostgreSQL, Docker, Cypress, and Material UI • Created a RESTful API that allowed for seamless integration with other applications
- Implemented a Responsive Design via Material UI that allowed the application to be used on various devices with the same user experience
- Created automated end-to-end tests using Cypress

Software Developer at Climalytics LLC, Austin, TX

May 2019 — September 2019

- Developed on a mobile application using React Native that improved user experience and increased app downloads by 20%
- Created a long-distance climbing competition dashboard to measure a user's current distance climbed and current ranking, utilized across various climbing gyms
- Implemented automated testing with Jest to ensure code quality and reduce the number of bugs in production
- Redesigned the user dashboard to create a user-friendly experience with ease to access a user's session data

PROJECTS

climbharder.app — Climbing Training Platform

- Designed and built a full-stack climbing training application that generates personalized training programs and automatically adjusts them based on each user's goals, feedback, and progress



EDUCATION

Master of Science: Computer Science, University of Texas, San Antonio

August 2021 — May 2023

Bachelor of Science: Computer Science, University of Texas, San Antonio

August 2012 — December 2017

Concentration in Software Engineering and Minor in Information Systems